

Experienced ML Scientist specializing in generative models (GANs, diffusion) and bioimaging, with 6+ years of experience developing end-to-end ML pipelines for industrial-scale microscopy and virtual staining. Ph.D. research focused on uncertainty quantification and generalizability of ML models for high-throughput biological imaging. Proven ability to translate complex biological challenges into scalable ML solutions, collaborating with cross-functional teams and open-source communities. Passionate about advancing computational biology through cutting-edge deep learning.

EXPERIENCE

- **GSK Pharma R&D Bioimaging Analytics** Philadelphia, USA.
Nov 2020 - Dec 2024
University Worker (Ph.D) (Remote)
Collaboration as part of Ph.D research focused on the responsible use of ML-based virtual staining via generative modelling within GSK Pharma R&D Bioimaging Analytics. Implemented end-to-end (E2E) solution for processing, generating and evaluating virtual fluorescence microscopy from unstained bright-field images.
 - **Responsibilities:** Explore, design, present, develop and maintain E2E pipeline with SOTA probabilistic ML approaches for multimodal image translation applications such as virtual staining on large datasets (5 million + images). Collate and review terabytes of diverse data from across multiple internal sites/labs, to generate hypotheses. Design reliable and meaningful quantitative methods to evaluate the performance of these methods in a way that translates to business recommendations. Set up and conduct large-scale experiments to test hypotheses and drive product development. Cross-functional partnership with stakeholders within Bioimaging from different departments and time zones to define AI-based solutions to biological problems. Adapt ML methods typically used in natural image domains to work in specific bioimaging scenarios and high-performance computing environments.
 - **Achievements:** Designed E2E PyTorch pipeline leveraging diffusion models and GANs for virtual staining of microscopy images, reducing staining time from weeks to minutes and enabling longitudinal live-cell studies. Published at ISBI 2023[4]. Collaborated with biologists and stakeholders to define ML solutions for industrial bioimaging challenges, improving model generalizability and reducing required trained models by 40% for large-scale deployment. Published at Bioimaging Computing workshop at ECCV 2024 [5]. Developed novel evaluation metrics for generative models, uncovering insights into performance undetectable by traditional measures (e.g., FID). Cross-functional leadership: Partnered with global teams (US/UK) to process 5M+ images across diverse datasets, aligning ML outputs with biological relevance.
- **Deep Learning at Marine Biological Lab (DL@MBL) [6]** Massachusetts, USA.
June 2024 - September 2024
Teaching Associate
Recruited to provide hands-on expertise on generative modelling approaches to image translation and to support students in applying deep learning methods to their specific bioimaging problem.
 - **Responsibilities:** Develop a repository for students to explore generative modelling approaches to image translation for virtual nuclei and cytoplasm staining. Partner with industry colleagues from Chan Zuckerberg Biohub to deliver exercise sessions on regression vs generative modelling approaches for image translation tasks.
 - **Achievements:** Led workshops on generative models for 3D nuclei segmentation, mentoring biologists with no prior ML experience. Created an open-source repository for virtual staining using generative models, adopted by Chan Zuckerberg Biohub for training exercises.
- **The Alan Turing Institute** London, UK.
Feb 2024 - September 2024
Core Developer
Invited back to the Turing as a core developer of Scivision, an open-source computer vision package designed at the Turing.
 - **Responsibilities:** Core developer of Scivision [7] an open-source computer vision package developed at The Alan Turing Institute. Sourced new models and data sets from potential new partners, designed and developed code to improve automation, deployment, and testing frameworks.
 - **Achievements:** Enhanced Scivision, an open-source CV package, by implementing CI/CD pipelines and expanding its model catalogue for bioimaging applications. Presented Scivision at AIUK 2023, fostering collaborations with Google DeepMind and Microsoft.

- University of Birmingham** Birmingham, UK.
Teaching Associate *Sep 2021 - June 2024*
 - **Responsibilities:** Supporting and mentoring undergraduate and postgraduate students on Data Structures & Algorithms, Artificial Intelligence, Computer Vision & Imaging, Neural Computation, Visualisation and Software Engineering & Professional Practises modules.
 - **Achievements:** Designed Python/Slurm workshops for Computer Vision, Neural Computation, and AI modules, bridging ML theory with biological applications.
- The Alan Turing Institute** London, UK.
Enrichment Student *Jan 2022 - Jan 2023*
 - **Responsibilities:** Contribute ideas and code for improving Scivision functionality as well as further building the community of developers and users to improve the adoption of Scivision. Research and collaborate with other Turing Fellows and enrichment students to improve my Ph.D research.
 - **Achievements:** Enhanced Scivision, an open-source CV package, by expanding its model catalogue for bioimaging applications. Planned and facilitated an all-day workshop on Uncertainty Quantification for Generative Models at The Alan Turing Institute inviting world-leading guest speakers from Google DeepMind, Microsoft, and The Alan Turing Institute to facilitate collaboration and discussion of the latest developments within this field.
- Haleon** London, UK
Data Scientist *Jul. 2018 - Oct. 2020*
 Recruited into GSK Consumer Healthcare (now Haleon) Market Analytics group for Northern Europe to implement ML-based forecasting methodologies, utilize automation to improve data gathering processes and provide commercial insights for the leadership team.
 - **Responsibilities:** Research and implement suitable ML-based forecasting methodologies to improve forecasting accuracy. Improve operational efficiencies through automation of data extraction, modelling and visualization. Collaborate with cross-functional teams of Senior Brand Managers, Finance Directors, and Tech to drive commercial decision-making.
 - **Achievements:** Implemented E2E pipeline for data gathering, data processing, a deep learning-based approach to forecasting (LSTM), and visualization of results. This led to an increase in market forecast accuracy for brand teams leading to data-centric decision making and ultimately successful brand launches.
- GSK Tech** London, UK
Technical Associate *Jan. 2017 - May. 2018*
 Joined the Business & Technology GSK Future Leaders Program working on the Global ERP system role out across GSK specialising in system security and design.
 - **Responsibilities:** Lead SAP cybersecurity testing for global ERP systems, addressing critical vulnerabilities.
 - **Achievements:** Successfully deployed testing across global-system, identifying and importantly addressing vulnerabilities improving robustness and adherence to policy guidelines. Designed and implemented standard operating procedures (SOPs) for RPA. Transforming business problems into RPA solutions. Represented GSK at RPA conferences.

EDUCATION

- Ph.D. Computer Science** Nov. 2020 – Present (April. 2025)
University of Birmingham (Krull Lab) *Birmingham, UK.*
 - **Focus:** Thesis: *Uncertainty Quantification for ML-Based Virtual Staining in High-Throughput Screening* (GSK-funded, industrial-scale bioimaging focus).
 - **Advisors:** Dr. Alexander Krull [9], Prof. Iain Styles [10] & Dr. Minh Doan [11]
- M.Sc Business Analytics; Distinction** Sep. 2018 – Aug. 2020
Imperial College Business School *London, UK.*
 - **Focus:** Deep learning for image analysis, Machine Learning, Data Structures & Algorithms, Database Query Languages, Linear Algebra, Statistics, Finance Analytics, Network Analytics and Optimization & Decision Models.
- BA Business & Management; 1st Class Honors** Sep. 2011 – Aug. 2014
University of Portsmouth *Portsmouth, UK.*

SKILLS

- Pytorch, Tensorflow, CUDA, Weights & Biases, GANS, Diffusion Models, Probabilistic ML, Fluorescence Microscopy, 3D Segmentation, Virtual Staining, Python (Advanced), SQL, Cross-Functional Team Leadership, Open-Source Development.

AWARDS, WORKSHOPS & INVITED TALKS

- **Data Science in Production with Python [12]**: 2-day course from Cambridge Spark focused on productionising ML pipelines.
- **Uncertainty Quantification for Generative Models [13]**: Independently planned and facilitated an all-day workshop at The Alan Turing Institute inviting 4 guest speakers (Google DeepMind, Microsoft, The Alan Turing Institute) and a round table discussion of the latest developments within the field.
- **Evaluation of virtual stain models for high-throughput screening [14]**: Oral presentation of works at the International Symposium for Biological Imaging (ISBI) 2023

SELECTED PUBLICATIONS

1. **Samuel Tonks**, Chih Hsu, Steve Hood, Ryan Musso, Ceriden Hopely, Minh Doan, Erin Edwards, Alexander Krull, and Iain Styles. "Evaluation of virtual staining for high-throughput screenings". In 20th IEEE International Symposium on Biomedical Imaging. IEEE, 2023. [15] Demonstrated ML-driven efficiency gains for industrial microscopy workflows.
2. Alexander Krull, Hector Basevi, Benjamin Salmon, Andre Zeug, Franziska Müller, **Samuel Tonks**, Leela Muppala, and Aleš Leonardis. "Image denoising and the generative accumulation of photons." In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, pp. 1528-1537. 2024.[16]Addressed model generalizability in diverse biological datasets.
3. **Samuel Tonks**, Cuong Quoc Nguyen, Steve Hood, Ryan Musso, Ceridwen Hopely, Steve Titus, Minh Doan, Iain Styles, Alexander Krull "Can virtual staining for high-throughput screening generalize?" 2024. Bioimaging Compute Workshop Paper European Conference for Computer Vision (ECCV) in Milan 2024. [17]

HOBBIES & INTERESTS

- Long-distance runner (Paris Marathon April 2025!)
- Learning to DJ (vinyl only) at Point Blank Music School London.